A one-dimensional obstruction to the printed random-polytope central limit assertion

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Scope. The source defines K_n as the convex hull of uniform points in a convex body $K \subset \mathbb{R}^d$, and asserts that the normalized volume deficit obeys a central limit theorem. Both printed versions omit a restriction $d \geq 2$. We take the usual positive-dimensional convex-body convention and the usual centering by the actual expectation and scaling by the actual standard deviation. The counterexample is the full-dimensional body $K = [0,1] \subset \mathbb{R}$, not a lower-dimensional subset of a higher-dimensional space. Only the CLT conjunct is refuted. No value is assigned to the source's unspecified sphere constants, and no contradiction to a corrected $d \geq 2$ theorem is claimed.

Theorem 1. *Let $n \geq 2$, let $X_1, \ldots, X_n$ be independent uniform samples in $[0,1]$, and let $Y_n = \mathrm{vol}(K) - \mathrm{vol}(K_n)$. Then*

$$\mathbb{E}Y_n = \frac{2}{n+1}, \qquad \mathbb{E}Y_n^2 = \frac{6}{(n+1)(n+2)}, \qquad \mathrm{Var}(Y_n) = \frac{2(n-1)}{(n+1)^2(n+2)} > 0.$$

The actual standardized laws of $Z_n = (Y_n - \mathbb{E}Y_n)/\sqrt{\mathrm{Var}(Y_n)}$ do not converge weakly to $N(0,1)$; their CDFs also fail the ordinary Gaussian CLT criterion.

The true sampling model and hull. The measure $U = \lambda|_{[0,1]}$ is already normalized, since $\lambda([0,1]) = 1$. The sample measure is the genuine finite product $U^{\otimes n}$, with coordinate projections as the random variables. Thus the coordinates are independent and have the normalized restriction law. The interval is compact, convex and has nonempty interior, and $\dim_{\mathbb{R}} \mathbb{R} = 1$. Put $A = \min_i X_i$, $M = \max_i X_i$ and $B = 1 - M$. For every finite sample the true convex hull is $[A, M]$: all points lie in this interval, and both endpoints are sample points. Its Lebesgue volume is $M - A$. Consequently $Y_n = A + B$, with $A, B \in [0,1]$ almost surely.

Moments derived from the product measure. For $u, v \in [0,1]$, the coordinate rectangle gives exactly

$$\mathbb{P}(A \geq u, B \geq v) = \big(\max(1-u-v, 0)\big)^n, \qquad \mathbb{P}(A \geq u) = \mathbb{P}(B \geq u) = (1-u)^n.$$

The joint formula includes the genuinely empty rectangle when $u + v > 1$; it is proved from the measure of that rectangle, not postulated as an order-statistic distribution.

For a measurable $X \in [0,1]$ and bounded measurable W on any finite measure space, Fubini applied to $\mathbf{1}_{t \leq X(\omega)} W(\omega) t^k$ proves

$$\int W \frac{X^{k+1}}{k+1} = \int_0^1 t^k \left(\int_{\{X \geq t\}} W \right) dt.$$

The Lean proof first verifies the pointwise cut integral $\int_0^1 \mathbf{1}_{t \leq a} t^k dt = a^{k+1}/(k+1)$ and integrability on the actual product measure. Taking $W = 1$, $k = 0, 1$ and using the marginal tails yields

$$\mathbb{E}A = \mathbb{E}B = \frac{1}{n+1}, \qquad \mathbb{E}A^2 = \mathbb{E}B^2 = \frac{2}{(n+1)(n+2)}.$$

Applying the same identity with $X = A, W = B$, then to B under the restricted measure on $\{A \geq u\}$, yields the true cross moment

$$\begin{aligned} \mathbb{E}(AB) &= \int_0^1 \int_0^1 \big(\max(1-u-v, 0)\big)^n \, dv \, du \\ &= \int_0^1 \frac{(1-u)^{n+1}}{n+1} \, du = \frac{1}{(n+1)(n+2)}. \end{aligned}$$

Thus expanding $(A+B)^2$ proves the two displayed moments of Y_n. The native variance identity $\mathrm{Var}(Y_n) = \mathbb{E}(Y_n^2) - (\mathbb{E}Y_n)^2$ gives the displayed positive variance. All these quantities are integrals of the actual hull deficit.

An obstruction at a Gaussian continuity point. Nonnegativity of Y_n and the actual moment formulas imply

$$\frac{\mathbb{E}Y_n}{\sqrt{\mathrm{Var}(Y_n)}} = \sqrt{\frac{2(n+2)}{n-1}} \leq \sqrt{8} < 3, \qquad Z_n \geq -3 \quad \text{almost surely.}$$

In particular, every actual pushforward law ν_n satisfies $\nu_n((-\infty, -4]) = 0$. The genuine standard Gaussian measure γ has $\gamma((-\infty, -4]) > 0$, because its density is strictly positive and $[-5, -4]$ has positive Lebesgue measure. Its singleton boundary mass at -4 is zero. The native Portmanteau theorem would therefore force $\nu_n((-\infty, -4]) \to \gamma((-\infty, -4])$ under weak convergence, an impossibility. The same point directly contradicts convergence of the real-valued CDFs. This proves both failures in the theorem.

Formalization and reproduction. The Lean family uses $n = k + 2$ with $k \in \mathbb{N}$, a cofinal indexing of all sample sizes at least two. In `Main.lean`, `polytope` is the native convex hull; `deficit` uses its native Lebesgue measure. The weighted-tail theorem proves the Fubini identity; the endpoint and deficit moment theorems compute the actual integrals and native variance. `standardizedLaw` is the native probability-measure pushforward, and the final weak-convergence and CDF theorems prove the convergence obstructions. There is no assumed density identification or external geometric identification. Literature is used only for the customary model and normalization, not as a proof of this counterexample.

The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

From `lean/`, run `lean -version`, then `./verify.sh`. The script runs the following build, exhaustive ordinary axiom check and fresh check:

```
lake build +Main
lake env lean -DwarningAsError=true Check.lean
lake env leanchecker --fresh Main
```

Only `propext`, `Classical.choice` and `Quot.sound` are allowed. No admitted proof, native evaluator proof or additional axiom is used.

References

[1] V. Vu, *Central limit theorems for random polytopes in a smooth convex set*, arXiv:math/0503559v1 (2005), pp. 1–2 (model and standardized CLT), p. 4 (variance order). `https://arxiv.org/abs/math/0503559`.

[2] Y.-C. Chen, *Order statistics from a continuous univariate distribution*, UW STAT 512, Lecture 9 (2022 Autumn), uniform-distribution case. `https://faculty.washington.edu/yenchic/20A_stat512/Lec9_Order.html`.